Lextract: LLM-Based Extraction of Relevant Market Definitions from European Commission Decisions

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Abstract. Relevant market definitions are central to merger and antitrust analysis, but collecting them at scale from European Commission decisions is difficult because the decisions are numerous and heterogeneous in structure and wording. We present Lextract, an open-source pipeline that filters decision PDFs, uses Gemini 2.0 Flash in two calls to locate the market-definition section and to isolate individual definitions, and stores the results as JSON. Applied to 600 cases, it produced 4,105 records that feed JurisMercatus, a semantic-search interface. We document the prompts, settings, exclusion rules and thresholds, give a worked example from source text to record, and combine the model with three symbolic quality checks, namely a structural validity test derived from 50 manually checked definitions, a source-fidelity test that checks whether each record occurs verbatim in its decision, and a manual inspection of 100 records. Of the records, 76.6% pass the structural test. Against all 699 retrievable decisions, 79% of the 4,067 verifiable records are verbatim and 9% near-verbatim. The fidelity test flags the remaining 12% as absent from their source, three quarters of them from short simplified-procedure decisions without market analysis, and removes them in a cleaned release of 3,607 records. The structural test accepts 71% of the flagged records, which shows that checking output against the source text adds information that resemblance-based checks miss. Annotation sheets and baseline scripts for a gold-standard evaluation are provided.

Keywords. legal information extraction, competition law, relevant market definition, large language models, quality control, European Commission

1. Introduction

Relevant market definition is a basic step in merger and antitrust analysis because it fixes the products and the geographic area within which market power is assessed [1]. Practitioners and researchers therefore need to know how the European Commission has delineated a given market before, for instance whether retail distribution of cars is national or EEA-wide, whether a product market was segmented or "left open", and how such findings changed over time. This is a concrete user problem. The Commission has published more than 6,000 merger and antitrust decisions [2] and adds roughly 280 a year [3]. The market-definition discussion sits under different headings and in different wording, so keyword search over PDFs is unreliable and manual collection is slow and hard to repro-

duce. The only dedicated resource we know of, the Caselex Market Definitions Module [4], is a paid, editorially curated web product without bulk or programmatic access.

This paper presents *Lextract*,[1] an open-source pipeline that extracts market-definition passages from Commission decision PDFs with a large language model (LLM) and stores them as structured records, and the 4,105-record collection behind *JurisMercatus*,[2] a semantic-search interface. The design follows the hybrid view that LLMs are good at locating and copying legally relevant passages and that their output should be checked by deterministic means. The paper makes four contributions.

- A complete description of the pipeline, covering prompts, model and settings, exclusion phrases and thresholds, with a worked example from source text to record (Sections 3–4).
- A description of the released collection and of what researchers can do with it (Section 5).
- A three-part evaluation combining a structural validity test, a *source-fidelity* test that checks whether each record occurs verbatim in its decision, and a manual inspection of 100 records (Sections 6–7).
- An analysis of the records the fidelity test flags, which traces them to short-form decisions and to example passages in the prompt, together with the corresponding fixes (Section 8) and the tools for a gold-standard evaluation (Section 9).

2. Related Work and Existing Resources

Legal information extraction combines deterministic, statistical and, more recently, generative methods. Agnoloni et al. presented the BO-ECLI Parser for extracting and linking legal references from European case law [5], and Darji et al. released annotated German legal-reference data and stressed the role of domain-specific gold data [6]. Recent work uses LLMs for semantic structuring. Janatian et al. turned legislation into structured pathways for expert systems [7]; Costa et al. extracted elements from Brazilian judicial decisions, evaluated them with precision and recall against expert annotations and published them with provenance [8]; Spada et al. extracted events and dates from EU procurement notices with expert validation [9]; others address data scarcity and context [10,11,12]. Lextract shares their pipeline view, targets complete market-definition passages and emphasises inexpensive deterministic checks of LLM output, which the hallucination literature motivates [13].

Relation to existing resources. Caselex [4] offers human-reviewed market definitions across many authorities and languages through a subscription and a web interface. Lextract complements it with a free, reproducible and updatable collection that offers programmatic access. The pipeline can be re-run on new decisions, every record carries case metadata and the decision URL, and an export tool writes CSV, JSONL, Parquet, SQLite, BibTeX and RIS. To our knowledge no openly licensed, machine-readable collection of Commission market definitions existed before.

[1] `https://github.com/shriyanyamali/Lextract/tree/lda2026`
[2] `https://jurismercatus.shriyanyamali.com`

3. The Lextract Pipeline

Lextract is a sequence of six Python scripts driven by `run_pipeline.py`. The prompts are reproduced verbatim in the repository (`docs/PROMPTS.md`) and summarised here.

3.1. Document Collection and Filtering

Step 1 reads the Excel export of the Commission's competition case-search portal and extracts, per case, the decision URL, case number, year and policy area, the last inferred from the URL path. *Step 2* downloads each PDF and extracts its text with PyPDF2. A document is excluded if it has exactly three pages and contains one of five phrases from the closing recital of short-form decisions adopted under the simplified procedure [14], namely "For the reasons set out in the Notice on a simplified", "European Commission has decided not to oppose the notified operation", "declare it compatible with the internal market", "This decision is adopted in application of Article 6(1)(b)" and "Merger Regulation and Article 57 of the EEA Agreement". Such clearances contain no market analysis, so any "definition" extracted from them would be spurious. Included documents are written to batch files with a metadata header and split into two groups at 80,000 characters so that long decisions can be processed separately; no other length filter is applied. Version 1.1.0 of the code matches the phrases independently of whitespace and applies the rule to documents of at most three pages (Section 8).

3.2. Section Identification (LLM call 1)

Step 3 sends each batch file to Gemini 2.0 Flash (`gemini-2.0-flash` via the `google-generativeai` SDK) with default generation settings, that is, without temperature, top-p or output-length overrides, system instruction or JSON mode. The prompt asks the model to find the first occurrence of the words "market definition", to start one line before it, to continue past sub-headings and to stop at the next heading of the same level (e.g. from "IV. Market Definition" to "V. . . ."), excluding that heading, and to reproduce the metadata header unchanged. The output is the market-definition section as plain text.

3.3. Definition Extraction (LLM call 2)

Step 4 sends each section to the same model with a prompt that asks for a JSON array of objects with the fields `case_number`, `year`, `policy_area`, `link`, `topic` and `text`. It instructs the model to extract "the entirety of each individual market definition" as "a direct quote of the multiple paragraphs", to treat "left open" passages as definitions, to drop paragraph numbers, footnote markers and footnotes, to re-join words split by PDF extraction and to assign a short `topic`. It states that a definition "is always at least 4 sentences long, but never longer than 12 sentences" and contains four worked examples on detergents, chemical intermediates, flame retardants and trimethylolpropane. The two-stage design limits the second call to a few pages of relevant text instead of a full decision.

3.4. Cleaning and Merging

Step 5 strips Markdown code fences from the model's output and *Step 6* concatenates all files into `data/output.json`. No de-duplication, truncation or editing of the text is performed, so the released records are exactly what the model returned.

Table 1. Worked example from Case M.9324. The upper part shows the PyPDF2 text of the decision, with footnotes and later paragraphs omitted, and the lower part shows the record produced by Lextract.

Source text (decision PDF, Section 4)

```
4. PRODUCT MARKET DEFINITION

4.1.1. Wholesale distribution of ICT and CE products

(12) In the precedents listed at paragraphs (13)-(15), the Commission
assessed the possible product markets for wholesale distribution of IT
products, telecommunication equipment and CE.

(13) In Avnet/Magirus,8 Tech Data/Scribona,9 Tech Data/Avnet,10 the
Commission has defined a market for the wholesale distribution of
IT products, but left open the question whether the market should be
further sub -segmented (i) by product categories (e.g. server, storage,
PCs, etc.), (ii) by direct and indirect sales11 or (iii) by different
distribution models (broadline vs value -added distribution ("VAD")).12

(14) In Argues/SHC13 and ALSO/PCF,14 the Commission identified a
separate market [...]
```

Extracted record


```
{"case_number": "M.9324", "year": "2019", "policy_area": "Merger",
 "link": "https://ec.europa.eu/competition/mergers/cases1/202048/m9324_
479_3.pdf",
 "topic": "Wholesale distribution of IT products",
 "text": "In Avnet/Magirus, Tech Data/Scribona , Tech Data/Avnet ,
the Commission has defined a market for the wholesale distribution of
IT products, but left open the question whether the market should be
further sub -segmented (i) by product categories (e.g. server, storage,
PCs, etc.), (ii) by direct and indirect sales or (iii) by different
distribution models (broadline vs value -added distribution ("VAD"))."}
```


4. Worked Example

Table 1 follows one passage through the pipeline. The source is Case M.9324 *ALSO / ABC Data Group* (12 June 2019), a 16-page Article 6(1)(b) decision. Call 1 located the heading “4. PRODUCT MARKET DEFINITION” and returned Section 4 up to, but excluding, “5. COMPETITIVE ASSESSMENT”. Call 2 returned seven records, and the one shown is paragraph (13). The model removed the paragraph number and footnote markers 8–12, kept the wording otherwise intact, including the extraction artefact “sub -segmented”, and labelled the topic. The structural test accepts the record (63 words, learned phrase “the commission has”, marker “left open”), and the fidelity test finds it verbatim in the decision (coverage 1.0). The sibling record for paragraph (14) is near-verbatim (coverage 0.83) because in the PyPDF2 text that paragraph is interrupted by a page-break footnote block, which the model correctly skipped. The other five records for this decision are verbatim.

Table 2. Records per year and structural validity rate.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	All
Records	9	66	621	492	699	594	691	618	297	18	4,105
Valid	3	45	467	421	551	477	509	472	184	14	3,143
Rate (%)	33	68	75	86	79	80	74	76	62	78	76.6

5. The JurisMercatus Collection

The collection (`data/jurismercatus_definitions.json` in the repository) contains 4,105 records from 600 cases (578 merger, 22 antitrust) and 704 distinct decision documents dated 2016–2025 (Table 2). The documents were exported in English from the case-search portal, and the released links identify each processed document. Cases contribute 1 to 45 records (median 6), records range from 9 to 1,182 words (median 67, mean 91), and the `topic` field holds 3,250 distinct free-text labels. A cleaned release (`data/jurismercatus_definitions_clean.json`, 3,607 records, 455 cases) omits the records that the fidelity test (Section 7) finds absent from their source and all records from documents of at most three pages.

What researchers can do with it. Each record pairs a passage with its case number, year, policy area and decision URL, so it can be cited and checked. Through JurisMercatus, a query such as "retail distribution of new passenger cars, geographic scope" returns the passages in which the Commission discussed that question across cases, ranked by embedding similarity. Exported to a spreadsheet or SQLite database, the records support questions such as how often the Commission leaves a definition open by sector and year, or how geographic findings (national, EEA-wide, worldwide) are distributed.

6. Evaluation Design

6.1. Reference Set and Structural Validity Test

We evaluate the collection with three complementary instruments, each of which measures a different property. The reference set (`evaluation/reference.json`) holds 50 records that the author selected from the pipeline output and checked against the source PDFs. They cover 39 merger cases, mostly from 2022–2024, and range from 33 to 557 words (median 135).

The structural test (`evaluate.py`) derives three checks from the reference set. The length check requires 26 to 668 words, the reference range widened by 20% on each side. The phrase check requires at least one of the 121 word n-grams ($n = 2$–5) that occur in at least 15% of reference records, such as "can be left open" or "for the purposes of this decision". The marker check requires at least one of 18 fixed markers, namely "relevant market", "product market", "geographic market", "market definition", "left open", "EEA-wide", "EEA wide", "national market", "Commission considers", "Commission found", "Commission has previously", "Notifying Party", "internal market", "serious doubts", "plausible market", "exact scope", "for the purposes of" and "cannot be excluded". Seven of the 50 reference records fail the marker check themselves, so the markers form a strict proxy. The test measures whether a record resembles a verified market-

definition passage in length and phrasing. Whether the text occurs in the decision is measured by the fidelity test, while passage boundaries and missed definitions are left to the gold-standard study (Section 9). We therefore report the result as a *structural validity rate*.

6.2. Source-Fidelity Test

The fidelity test (`verify_fidelity.py`) asks whether a record is copied from its decision. Record and PyPDF2 decision text are normalised to lower-case letters only, which removes whitespace, digits and punctuation and so neutralises paragraph numbers, footnote markers and split-word artefacts. The record is cut into 40-letter windows at stride 20, and *coverage* is the fraction of windows found in the source. Records with coverage of at least 0.95 count as *verbatim*, from 0.80 to 0.95 as *near-verbatim*, from 0.50 to 0.80 as *partial* and below 0.50 as *absent*. Whether a copied passage is the right one is left to the gold-standard study. The test was run in October 2026 on every record whose decision could be fetched from the stored URL, covering 699 of the 704 documents and 4,067 of the 4,105 records; four URLs of case M.11241 now redirect and one points to an HTML index page.

6.3. Manual Inspection

To interpret the structural test's verdicts, 50 passing and 50 failing records were drawn at random (seeded) and read. Each was labelled from the record text alone. Label A1 marks a market-definition statement, such as a precedent, a party submission, a market-investigation finding or a Commission conclusion on product or geographic scope. Label A2 marks reasoning within a market-definition analysis without a definitional claim, B a fragment such as a bare heading, and C text outside market definition, such as competitive assessment, remedies, party descriptions or legal background. The labels are released with the sample (`evaluation/manual_inspection_sample.csv`). This single-annotator reading checks face validity. Borderline cases include a conclusion that a transaction raises no serious doubts "regardless of the exact product market definition", labelled C, and a list of "affected markets", labelled A1. Precision, recall and F1 require the full list of definition passages in whole decisions, and the choice of annotation unit (paragraph, sub-section or conclusion) affects the scores. The repository includes an annotation protocol and a script that produces, for N random decisions, a sheet listing every record next to the decision text with columns for true positive, boundary error, false positive and missed definition (Section 9).

7. Results

7.1. Structural Validity

Of the 4,105 records, 3,143 (76.6%) pass all three checks. The 962 failures divide into 668 marker only, 185 length only, 80 length and marker, 19 phrase and marker, and 10 all three. Of the 275 length failures, 261 are shorter than 26 words and 14 longer than 668. The yearly rate (Table 2) lies between 68% and 86% in every year with more than 50 records except 2024 (62%), and the 65 antitrust records reach 62%. Among the 777

Table 3. Source-fidelity test on all retrievable records and on the reference set's source documents (five URLs, 38 records, were unretrievable).

	Collection	Reference set
Documents checked (of which ≤3 pages)	699 (136)	38 (1)
Records checked	4,067	48
Verbatim (coverage ≥0.95)	3,221 (79.2%)	44
of which exact match (1.0)	2,985 (73.4%)	–
Near-verbatim (0.80–0.95)	349 (8.6%)	3
Partial (0.50–0.80)	26 (0.6%)	0
Absent (<0.50)	471 (11.6%)	1
Absent, from ≤3-page documents	353 of 380	1 of 1
Absent, from longer documents	118 of 3,687	0 of 47

records without a marker, 535 (69%) still contain the word "market", 105 "national" and 65 "EEA", so many express scope in wording the marker list does not cover, such as "the market is national in scope"; 242 (31%) do not mention a market.

7.2. Source Fidelity

Of the 4,067 verifiable records (Table 3), 3,221 (79.2%) are verbatim, 349 (8.6%) near-verbatim, 26 (0.6%) partial and 471 (11.6%) absent. Absent records concentrate in documents without market analysis. Of the 699 documents, 136 have at most three pages, and 353 of their 380 records are absent; the remaining 27 are copied boilerplate such as party descriptions. Of the 118 absent records from longer documents, 26 reproduce the prompt's example passages (Section 8), 32 come from twelve documents of four to seven pages, and 60 come from 19 longer documents. The latter are mainly ancillary decisions that refer back to a main clearance decision, whose records restate the main decision's definitions in words that do not occur in the document, and a few long decisions whose records share only fragments with the text. Of the 349 near-verbatim records, 292 have at most two missing windows, which corresponds to a single break. Inspection of such cases in twelve documents found two causes, footnote text interleaved at a page break in the PyPDF2 text, as in the worked example, and a skipped intermediate paragraph, such as a party's submission between the Commission's precedent and its conclusion in M.10745. No paraphrase was found.

Of the 48 reference records whose documents could be retrieved, 44 are verbatim, three near-verbatim and one absent. Of the 471 absent records, 336 (71%) pass the structural test, close to the rate for verbatim records (2,475 of 3,221, 77%), so structural resemblance does not indicate presence in the source.

7.3. Manual Inspection

Among the 50 passing records, 49 were labelled A1 and one C, a competitive-assessment conclusion. One of the 49 is a prompt example that the fidelity test flags (Section 8). Among the 50 failing records, 38 were labelled A1 and three A2, so 41 (82%, interval 69–90%) contain market-definition content. One is a bare heading (B), and eight (16%) fall outside market definition (C), covering remedies reasoning, a party descrip-

tion, Treaty text and competitive assessment. Seven of the 41 are complete one-sentence scope statements below the 26-word minimum, such as "For AIs, the Commission therefore takes the view that the markets are at least EEA-wide", and four of them fail on length alone. One is a coherent 727-word party submission that exceeds only the maximum. Most structural failures therefore stem from wording and length, and fidelity is measured separately (Section 7).

8. Error Analysis and Fixes

Short-form decisions. Most absent records, 353 of 471, come from simplified-procedure clearances. These contain the exclusion phrases, but PyPDF2 renders them with inserted spaces and line breaks, as in "on a simpli fied procedure", so exact matching does not find them. In a 40-document pilot sample, none of eight three-page documents matched a phrase, and a two-page document that matched fell outside the three-page rule. Both prompts presuppose a market-definition section, so for a decision without market analysis the model produced precedent-style passages, and eight records reproduce the simplified-procedure recital itself. Version 1.1.0 matches the phrases independently of whitespace and applies the rule to documents of at most three pages, which excludes all nine short-form documents of the pilot. The fidelity script removes absent records and, optionally, all records from documents of at most three pages, which produced the cleaned release.

Prompt examples. The four example passages in the extraction prompt appear in 30 records across 11 cases that are not their source. The structural test accepts them, and the fidelity test flags all 30. The next version of the prompt omits the examples.

Boundaries, duplicates and wording. Near-verbatim records show the model splicing across footnotes and occasionally omitting a paragraph. At the other end, 261 records are shorter than 26 words, mostly single conclusion sentences, and 14 exceed 668 words. In addition, 117 records repeat the text of another record, mostly one passage stored under two topic labels. These segmentation questions are part of the gold-standard study. Most remaining structural failures are genuine definitions in wording the 18 markers do not cover, notably "national in scope".

9. Limitations and Future Work

Gold standard. Precision, recall and F1 against an independently annotated set of all definitions in a decision remain to be measured. The fidelity test establishes that 88% of records are copied from their source; whether they are the right passages and whether any were missed is the subject of a planned annotation of ten decisions, for which `make_annotation_sheet.py` provides the sheet and protocol. *Baselines.* The two-stage design and the choice of Gemini 2.0 Flash, taken for its free-tier rate limits and long context, have yet to be compared with a single-call variant, a heading-based rule baseline and other LLMs. The first two are implemented in `baselines/` and feed the same evaluation scripts. *Annotation.* The manual inspection and the reference set were each labelled once, without a second annotator. *Scope.* The pipeline processes English-language decisions of one authority, and its output depends on the quality of the PDF text.

10. Conclusion

Lextract turns heterogeneous Commission decisions into 4,105 structured market-definition records with a two-call LLM pipeline whose prompts, settings and filters are documented in full. Across the collection, 79% of records are verbatim copies of their source and a further 9% are near-verbatim. The source-fidelity test identifies the remaining records, most of them from short decisions without market analysis, which a resemblance-based check accepts, and produces a cleaned release of 3,607 records. Combining LLM extraction with a deterministic check against the source text is inexpensive and applies to any extraction pipeline over legal documents.

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References

[1] European Commission. Commission Notice on the definition of the relevant market for the purposes of Union competition law. Official Journal of the European Union; 2024. OJ C, C/2024/1645, 22.2.2024.

[2] Bernhardt L, Dewenter R. The Impact of the More Economic Approach on EU Merger Decisions. Stanford Computational Antitrust. 2024. Available from: https://law.stanford.edu/wp-content/uploads/2024/12/Bernhardt-Dewenter.pdf.

[3] Affeldt P, Duso T, Szücs F. 25 Years of European Merger Control. International Journal of Industrial Organization. 2021;76:102720. Available from: https://doi.org/10.1016/j.ijindorg.2021.102720.

[4] Caselex. Caselex Market Definitions Module; 2026. Commercial product, https://www.caselex.eu/services/service.

[5] Agnoloni T, Bacci L, Peruginelli G, van Opijnen M, van den Oever J, Palmirani M, et al. Linking European Case Law: BO-ECLI Parser, an Open Framework for the Automatic Extraction of Legal Links. In: Legal Knowledge and Information Systems: JURIX 2017. vol. 302 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2017. p. 113-8. Available from: https://doi.org/10.3233/978-1-61499-838-9-113.

[6] Darji H, Mitrović J, Granitzer M. A Dataset of German Legal Reference Annotations. In: Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law. ICAIL '23. New York, NY, USA: Association for Computing Machinery; 2023. p. 392-6. Available from: https://doi.org/10.1145/3594536.3595173.

[7] Janatian S, Westermann H, Tan J, Savelka J, Benyekhlef K. From Text to Structure: Using Large Language Models to Support the Development of Legal Expert Systems. In: Legal Knowledge and Information Systems: JURIX 2023. vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 167-76. Available from: https://doi.org/10.3233/FAIA230962.

[8] Costa MZ, Robson DF, Vieira TBP, Bourguet JR, Guizzardi G, Almeida JPA. Automated Semantic Annotation Pipeline for Brazilian Judicial Decisions. In: Legal Knowledge and Information Systems: JURIX 2024. vol. 395 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2024. p. 226-38. Available from: https://doi.org/10.3233/FAIA241248.

[9] Spada I, Nai R, Audrito D, Trifiletti VSM, Sulis E. An Expert-Validated LLM Framework for Transforming Legal Procurement Texts into Actionable Data. In: Legal Knowledge and Information Systems: JURIX 2025. vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 356-63. Available from: https://doi.org/10.3233/FAIA251610.

[10] Nguyen PM, Nguyen HT, Zin MM, Satoh K. Data Augmented Pipeline for Legal Information Extraction and Reasoning. In: Proceedings of the Twentieth International Conference on Artificial Intelligence and Law. ICAIL '25. New York, NY, USA: Association for Computing Machinery; 2025. p. 481-2. Available from: https://doi.org/10.1145/3769126.3769200.

[11] Breton J, Billami MM, Chevalier M, Nguyen HT, Satoh K, Trojahn C, et al. Leveraging LLMs for legal terms extraction with limited annotated data. Artificial Intelligence and Law. 2025. Available from: https://doi.org/10.1007/s10506-025-09448-8.

[12] Castano S, Ferrara A, Furiosi E, Montanelli S, Picascia S, Riva D, et al. Enforcing legal information extraction through context-aware techniques: The ASKE approach. Computer Law & Security Review. 2024;52:105903. Available from: https://doi.org/10.1016/j.clsr.2023.105903.

[13] Kalai AT, Nachum O, Vempala SS, Zhang E. Why Language Models Hallucinate. arXiv; 2025. Available from: https://arxiv.org/abs/2509.04664.

[14] European Commission. Commission Notice on a simplified procedure for treatment of certain concentrations under Council Regulation (EC) No 139/2004. Official Journal of the European Union; 2013. OJ C 366, 14.12.2013, p. 5.